Basis of Record Analysis

Sweden - GBIF Data Gap Assessment

February 13, 2026

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Overview

This report analyses **how different types of biodiversity observations** contribute to GBIF data coverage in Sweden. Understanding the composition of the data by basis of record is essential for identifying biases and planning targeted data mobilisation efforts.

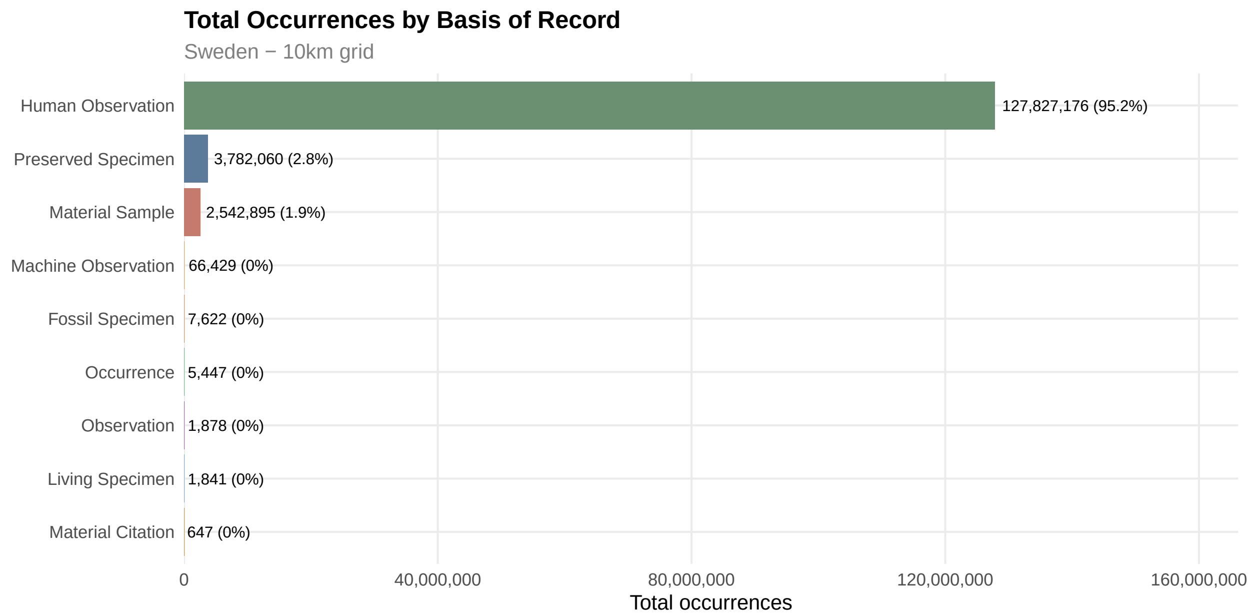
GBIF distinguishes several basis-of-record categories, each representing a different method of documenting biodiversity:

- **Human Observation** - direct field sightings (e.g. citizen science, bird surveys)

- **Preserved Specimen** - museum and herbarium collections
- **Machine Observation** - camera traps, acoustic sensors, remote sensing
- **Material Sample** - DNA, tissue, environmental samples
- **Fossil Specimen** - palaeontological records
- **Living Specimen** - botanical gardens, zoos

Occurrence Volume by Basis of Record

Total Occurrences

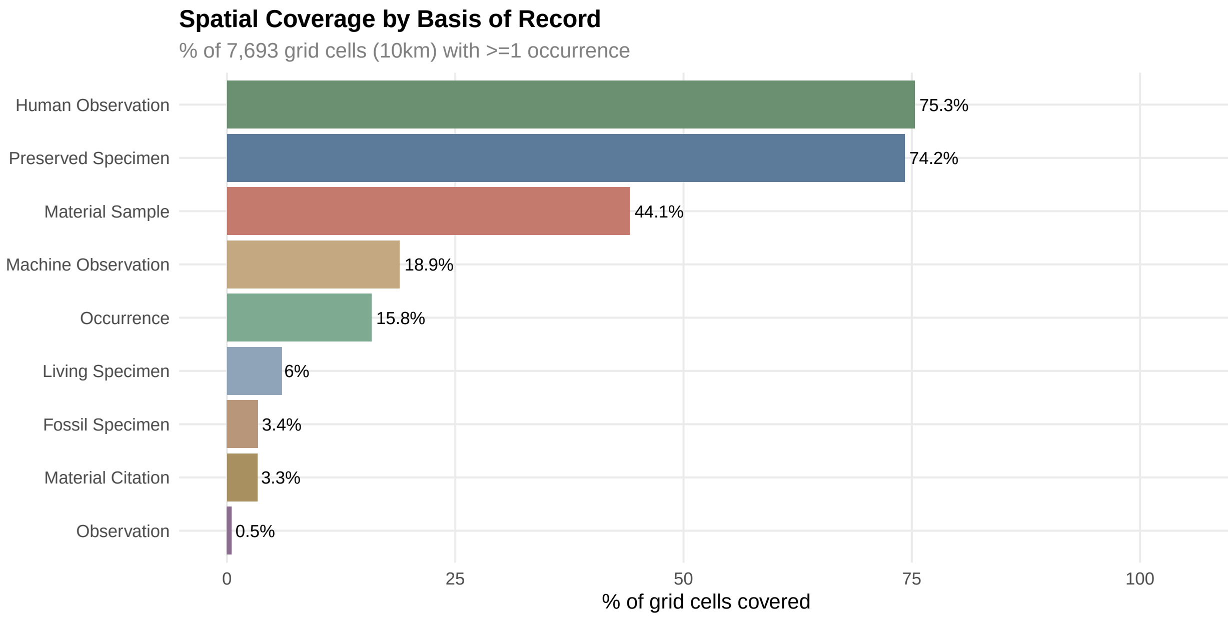


Summary Table

Basis of Record	Total Occurrences	% of All	Cells with Data	Spatial Coverage (%)	Species Detections
Human Observation	127,827,176	95.2	5,791	75.3	5,353,102
Preserved Specimen	3,782,060	2.8	5,711	74.2	1,756,824
Material Sample	2,542,895	1.9	3,391	44.1	221,428
Machine Observation	66,429	0.0	1,451	18.9	3,176
Fossil Specimen	7,622	0.0	263	3.4	1,041
Occurrence	5,447	0.0	1,212	15.8	3,212
Observation	1,878	0.0	38	0.5	1,568
Living Specimen	1,841	0.0	463	6.0	1,381
Material Citation	647	0.0	253	3.3	437

Spatial Coverage

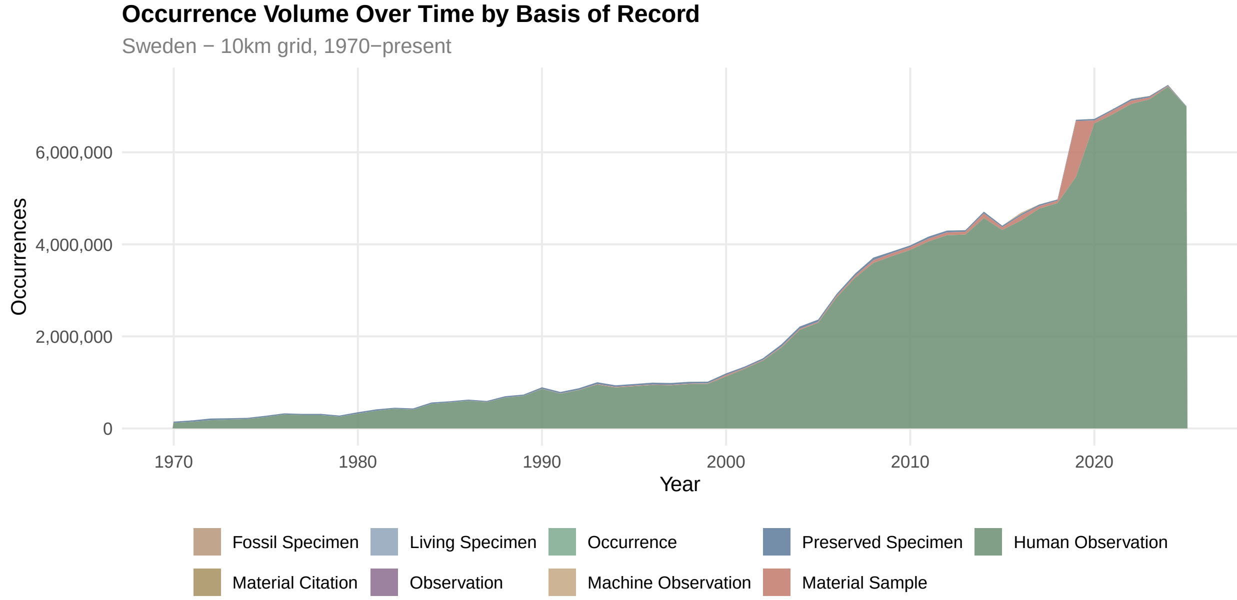
How well does each basis of record cover the spatial grid? A basis type that covers many cells provides broad geographic information, while one concentrated in few cells may reflect institutional or logistical biases.



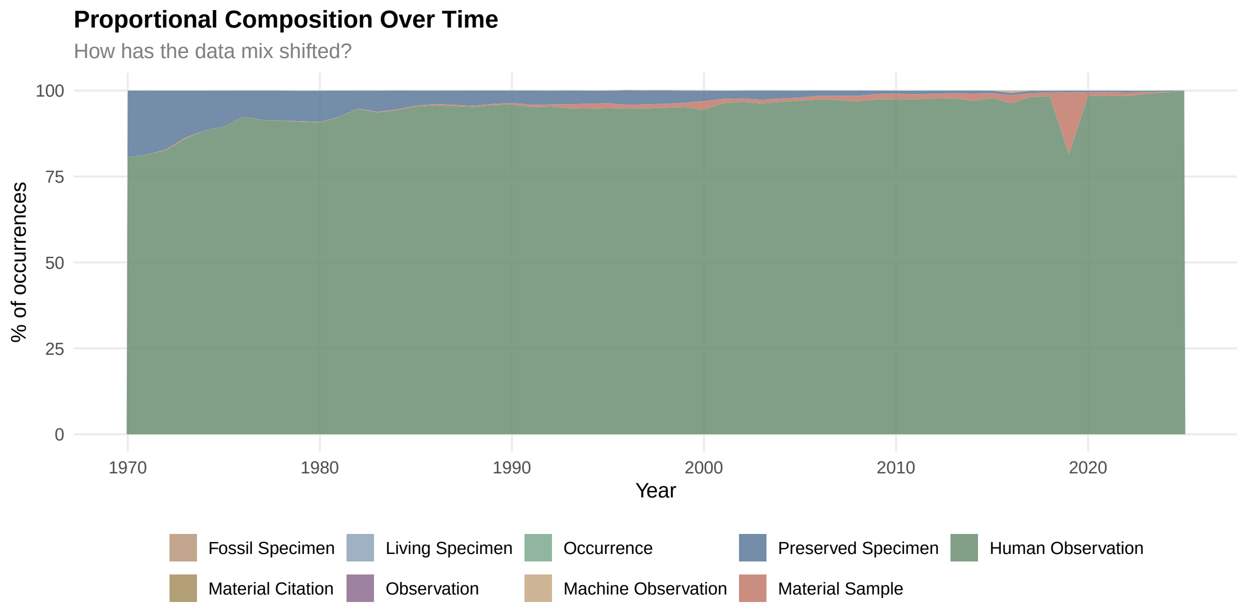
Temporal Trends

How has each basis of record evolved over time? Shifts in the dominant observation type reflect changes in biodiversity monitoring practice - from museum collecting to citizen science platforms.

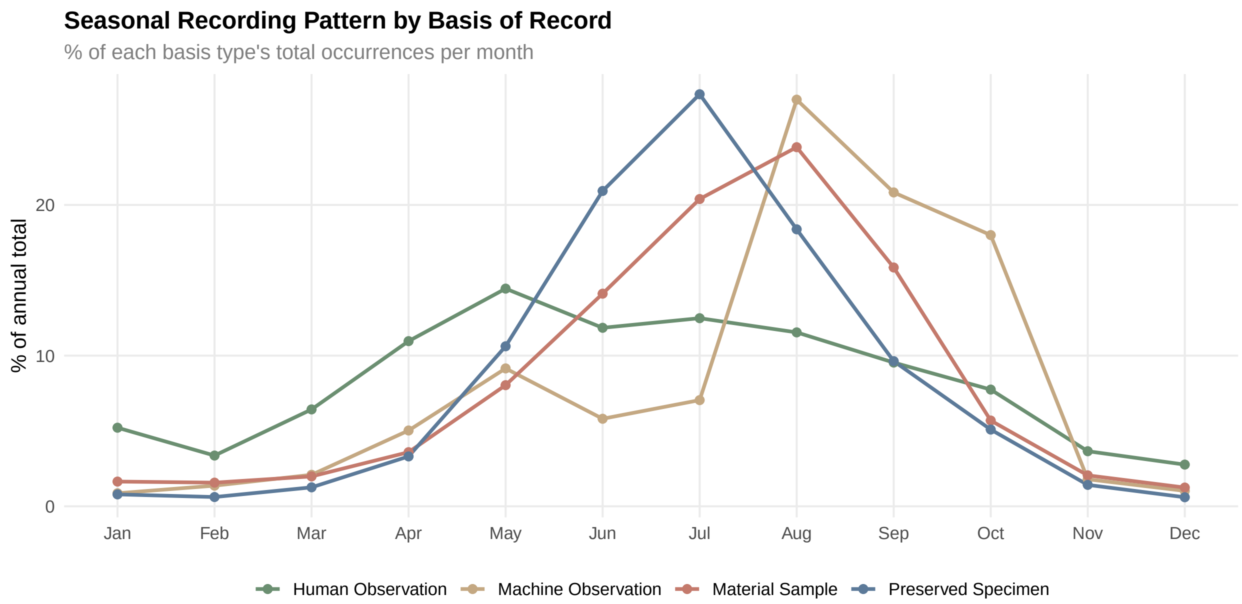
Annual Trend (Stacked Area)



Annual Trend (Proportional)

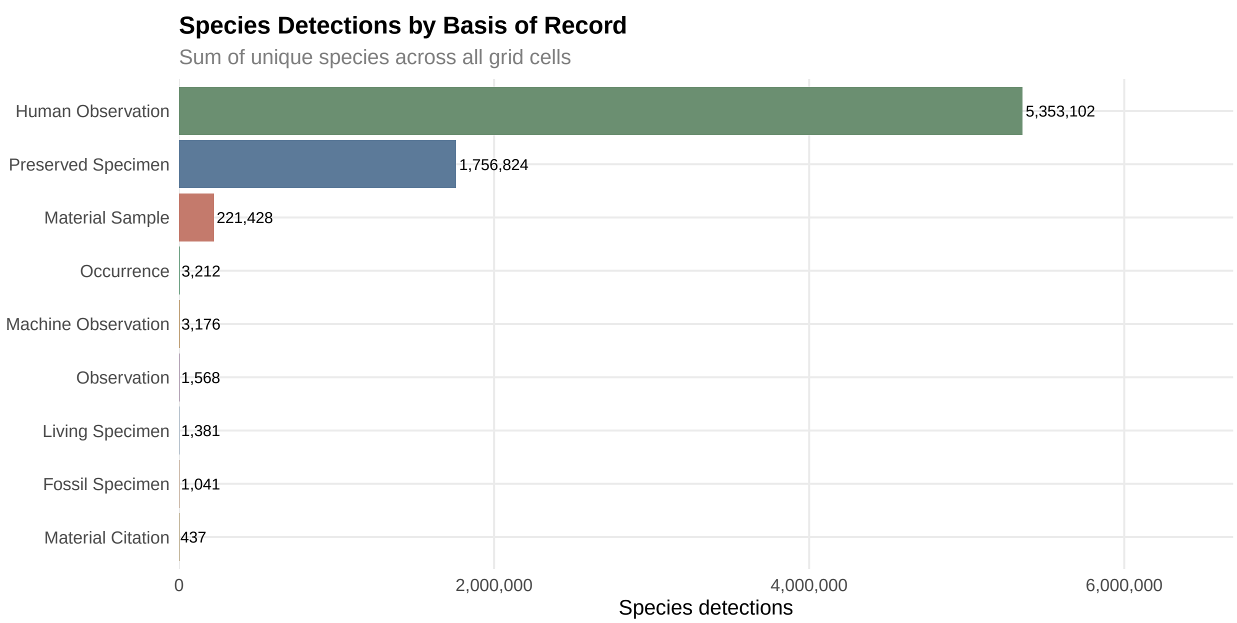


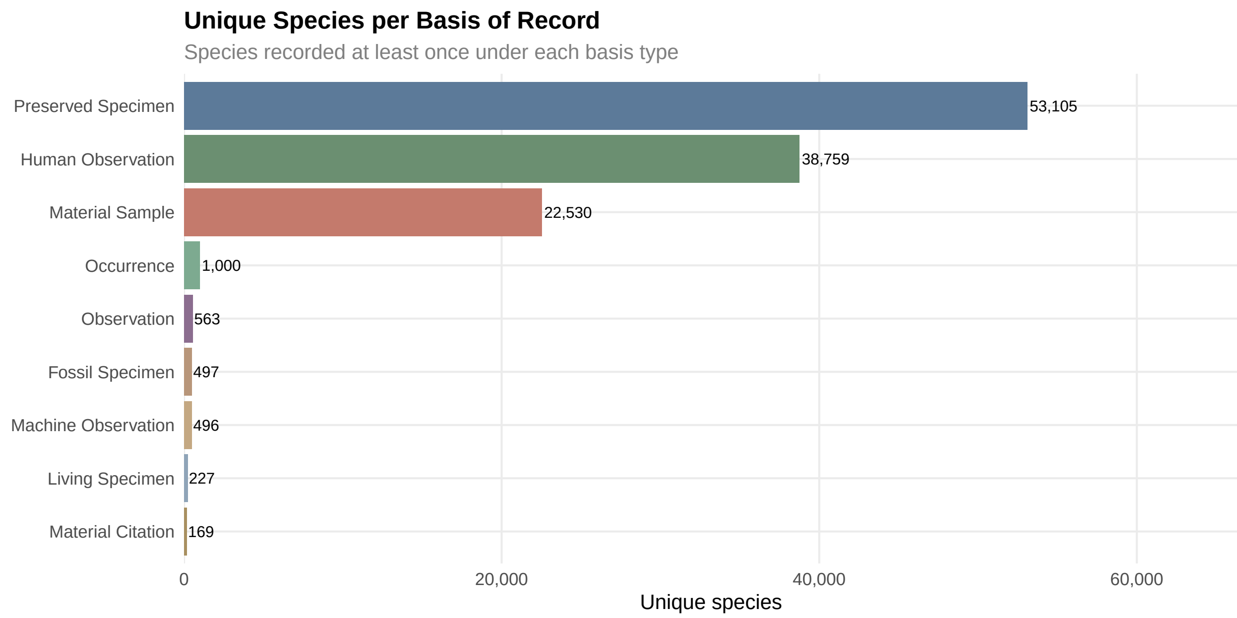
Seasonal Patterns



Species Coverage

Which basis types record the most species? And how unique is each - do preserved specimens capture species missed by human observations?





Taxonomic Coverage by Basis

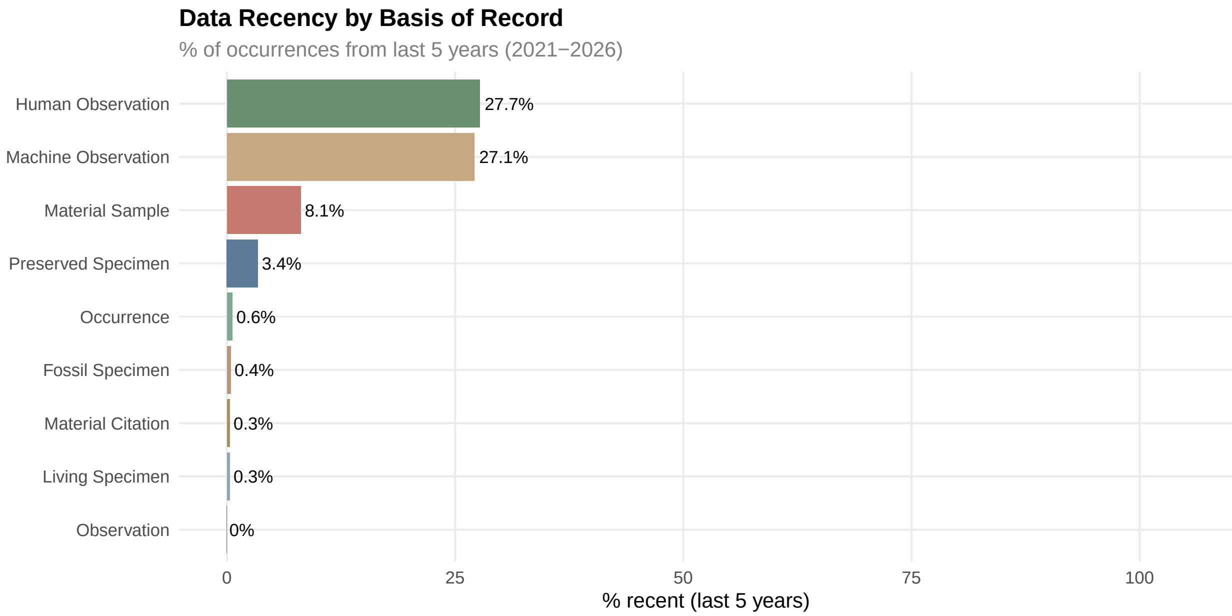
How well does each basis of record cover the national taxonomic backbone? This shows whether museum specimens and citizen science records document different parts of the tree of life.

Complementarity Analysis

Different basis types often complement each other. Preserved specimens may cover taxa poorly represented in citizen science data, and vice versa.

Basis A	Basis B	Species in A	Shared	Only in A	% of A shared
Human Observation	Preserved Specimen	38759	33705	5054	87.0
Material Sample	Preserved Specimen	22530	12976	9554	57.6
Preserved Specimen	Human Observation	53105	33705	19400	63.5
Material Sample	Human Observation	22530	11617	10913	51.6
Preserved Specimen	Material Sample	53105	12976	40129	24.4
Human Observation	Material Sample	38759	11617	27142	30.0

Data Quality Indicators



Basis of Record	Median Year	% Last 5 Years	% Last 10 Years
Human Observation	2015	27.7	48.3
Machine Observation	2016	27.1	90.6
Material Sample	2019	8.1	66.8
Preserved Specimen	1969	3.4	7.4
Occurrence	1947	0.6	1.5
Fossil Specimen	1922	0.4	0.4
Living Specimen	2001	0.3	6.9
Material Citation	2011	0.3	33.7
Observation	1995	0.0	0.0

Key Findings

- **Dominant data source**: Human Observation accounts for 95.2% of all occurrences (127,827,176 records)
- **Best spatial coverage**: Human Observation covers 75.3% of grid cells
- **Most current data**: Human Observation has 27.7% of records from the last 5 years
- **Most historical data**: Observation has a median year of 1995 (only 0% from the last 5 years)

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